SIH 26001 • JURY Q&A BRIEF

# SACHET: What It Does, Where It Falls Short for NER Landslides, and How We Close the Gap

*A short, evidence-backed answer for when the jury asks: "Doesn't SACHET already do this?"*

## 1. What SACHET Actually Is

SACHET ("Integrated Alert System") is India's national disaster-alert dissemination platform — built by C-DOT for NDMA, launched nationwide in August 2021, based on the international Common Alerting Protocol (CAP). It has delivered over 134 billion emergency messages to date.

| What it does | Detail |
|---|---|
| Alert-Generating Agencies (AGAs) | IMD, CWC, INCOIS, GSI, DGRE, FSI feed hazard alerts into SACHET |
| Alert-Authorising Agencies (AAAs) | SDMAs/State authorities approve and geo-target the alert |
| Dissemination channels | SMS, Cell Broadcast, Radio, TV, Siren, Social Media, Web Portal, App, Satellite (GAGAN/NavIC) |
| Coverage | All States & UTs, multilingual, geo-targeted |

**RESEARCH INSIGHT**

SACHET is a distribution pipe, not a prediction engine. It broadcasts whatever an Alert-Generating Agency hands it — it does not itself decide that a landslide is likely somewhere.

### How an alert actually reaches a citizen

*AGA (e.g. GSI) detects/forecasts a hazard → generates a CAP-format alert → SDMA/AAA authorises & geo-targets it → SACHET pushes it out via SMS / Cell Broadcast / App / Siren → citizen receives it (if in network range).*

## 2. Where the Gap Actually Is

**THE CENTRAL POINT**

GSI is already listed as an Alert-Generating Agency inside SACHET for landslides. The problem isn't SACHET's plumbing — it's that GSI's own landslide forecasting (its National Landslide Forecasting Centre, live since July 2024) currently issues forecasts only for Kalimpong, Darjeeling and the Nilgiris. For the NER, there is simply no landslide-specific alert being generated to hand to SACHET in the first place.

| # | Gap | Why It Exists |
|---|---|---|
| 1 | No landslide alert source for the NER | GSI's operational forecasting centre covers only 3 pilot districts nationally (Kalimpong, Darjeeling, Nilgiris) — the NER isn't among them |
| 2 | Regional forecasts, not slope/village-specific | Even where GSI's system runs, it forecasts risk at district/regional scale — it cannot say which specific slope or village will fail, only that the region is at risk |
| 3 | One-way broadcast only | SACHET pushes alerts out; it has no built-in channel for a citizen or field officer to report a crack, slope movement, or blocked road back in |
| 4 | Depends on mobile network towers | SMS/Cell Broadcast delivery fails in NER hill villages with no or patchy tower coverage — there is no offline/store-and-forward fallback |
| 5 | Text alert, no visual context | A CAP alert is a structured message — it isn't bundled with a live risk map or road-connectivity view a district officer could act on immediately |
| 6 | No explanation attached | Citizens receive a warning level but not why it was raised — this limits trust and can increase alert fatigue over time |

## 3. Where We Overcome Each Gap

We are not proposing to replace SACHET — it already works, is government-backed, and reaches billions of messages. We are proposing to be the missing piece upstream and downstream of it for the NER's landslide problem specifically.

| Gap | Our Response |
|---|---|
| No NER landslide alert source | Build the real-time NER-specific prediction engine (rainfall + soil moisture + satellite + terrain) that GSI does not yet run here — and feed its output into SACHET as a proper Alert-Generating input, exactly the role GSI already plays elsewhere |
| Regional, not slope-specific, forecasts | Add village/road-level resolution using local sensor data, DEM/slope data and citizen reports layered on top of the regional picture — sharper than a district-wide bulletin |
| One-way broadcast only | Add a geo-tagged citizen/field-officer reporting feature (photo of a crack or blocked road) — a two-way channel SACHET was never designed to have |
| Depends on mobile towers | Build an offline-first app with store-and-forward sync, so reports and alerts still move once any connectivity — even brief — becomes available |
| Text alert, no visual context | Pair every alert with a live GIS risk heatmap and road-connectivity dashboard for district disaster officials |
| No explanation attached | Show a simple "why this alert" panel — which data (rainfall spike, soil saturation, citizen report) triggered the risk level |

**WHAT THIS MEANS FOR US**

Positioning for the jury: "SACHET is the last mile for delivery. We are the missing first mile for prediction, and the missing return path for citizen input — both specific to NER landslides."

## 4. Summary — Gaps We Can Work On

| Gap in SACHET (for NER landslides) | Root Cause | What We Add |
|---|---|---|
| No landslide alerts issued for NER | GSI's forecasting centre covers only 3 districts nationally, none in NER | NER-specific real-time prediction engine feeding into SACHET |
| Forecasts are regional, not local | Regional rainfall-threshold model can't pinpoint a slope/village | Village/road-level risk scoring using sensors + terrain + citizen data |
| No citizen reporting channel | SACHET is built for one-way broadcast, not intake | Geo-tagged photo/video reporting app for cracks & blocked roads |
| Fails in low/no-network areas | Delivery relies on mobile towers with no offline fallback | Offline-first app with store-and-forward sync |
| No visual/GIS context with alerts | CAP messages are text-based, not map-based | Live GIS risk heatmap + road-connectivity dashboard |
| No explanation for citizens | Alerts state a risk level, not the reasoning | "Why this alert" panel showing the triggering data |

**KEY TAKEAWAY**

Every gap above is documented from GSI's and NDMA's own published scope — not assumed. This table is the one page to return to if a juror pushes on "why not just use SACHET."